Visualizing US Natural Disaster Declarations: Trends and Patterns

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1. Introduction

Natural disasters are significant events that can cause severe damage to communities, infrastructure, and the environment. Understanding historical disaster patterns can help governments and organizations prepare more effectively for future disasters. Data analysis and visualization play an important role in identifying trends and patterns in disaster occurrences.

This project focuses on analyzing disaster declarations in the United States using historical disaster data. The aim of the project is to explore patterns in disaster occurrences over time and across different geographic regions. By transforming raw data into visual insights, this project helps identify trends in disaster frequency, seasonal patterns, and regional variations.

Through the use of data visualization techniques, this project presents disaster trends using interactive dashboards that allow users to explore the data dynamically.

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2. Project Objectives

The main objectives of this project are:

- To clean and preprocess disaster declaration data for analysis
- To analyze temporal trends in disaster occurrences
- To identify seasonal patterns in disaster declarations
- To examine the geographical distribution of disasters across US states
- To develop interactive dashboards using Power BI
- To derive insights that can support disaster preparedness and planning

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3. Dataset Description

The dataset used in this project contains records of disaster declarations in the United States. Each record represents a disaster event officially declared by federal authorities.

The dataset includes information such as the disaster type, location, start date, end date, and assistance programs provided.

Key Attributes in the Dataset

Column Name	Description
Declaration Number	Unique identifier for each disaster declaration
Declaration Type	Type of declaration (Emergency or Major Disaster)
Declaration Date	Date when the disaster was officially declared
State	US state affected by the disaster
County	Specific county affected
Disaster Type	Category of disaster (Flood, Hurricane, Storm, etc.)
Disaster Title	Description of the disaster event
Start Date	Date when the disaster began
End Date	Date when the disaster ended
Close Date	Date when the disaster case was closed

This dataset provides valuable information for analyzing both temporal and geographical patterns of disasters in the United States.

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4. Data Cleaning and Preprocessing (Milestone 1)

Before performing analysis, the dataset required cleaning and preprocessing to ensure data quality and consistency.

Python was used for data preprocessing using the **Pandas library** in Jupyter Notebook.

Data Cleaning Steps

1. Handling Missing Values
Some columns contained missing values, particularly in fields such as county and close date.

These values were either handled appropriately or retained where they did not significantly affect the analysis.

2. Removing Duplicate Records
Duplicate entries were checked using the declaration number field to ensure each disaster record appeared only once.

3. Date Conversion
The start date and end date columns were converted into proper datetime format to enable time-based analysis.

4. Feature Engineering

Additional columns were created to simplify analysis:

- Year
- Month
- Month Name
- Disaster Duration (in days)

The disaster duration was calculated using the difference between the start date and end date.

Example formula:

Duration = End Date − Start Date

These preprocessing steps ensured that the dataset was structured properly for visualization and analysis.

```python
#Task -3 : Data Cleaning

#3.1 - Remove Exact Duplicates
df = df.drop_duplicates()
print("After removing exact duplicates:", df.shape)
```

```
After removing exact duplicates: (46185, 14)
```

```python
#3.2-Keep Unique Disaster Declarations
```

[redacted]

```python
df = df.drop_duplicates(subset=['Declaration Number'])
print("After keeping unique disasters:", df.shape)
```

```
After keeping unique disasters: (3427, 14)
```

```python
#3.3-Convert Date Columns
df['Start Date'] = pd.to_datetime(df['Start Date'], errors='coerce')
df['End Date'] = pd.to_datetime(df['End Date'], errors='coerce')
df['Close Date'] = pd.to_datetime(df['Close Date'], errors='coerce')

#3.4-Remove Missing Start Dates

df = df.dropna(subset=['Start Date'])

#3.5-Standardize Categorical Fields
df['Disaster Type'] = df['Disaster Type'].str.upper().str.strip()
df['State'] = df['State'].str.upper().str.strip()

#3.6-Feature Engineering (Extract Date Features)
df['Year'] = df['Start Date'].dt.year
df['Month'] = df['Start Date'].dt.month
df['Month_Name'] = df['Start Date'].dt.month_name()

#3.7 Create Duration Variable
df['Duration_Days'] = (df['End Date'] - df['Start Date']).dt.days
df = df[df['Duration_Days'] >= 0]
```

Figure 1: Data Cleaning Process in Python

5. Temporal Trend Analysis (Milestone 2)

Temporal analysis was performed to understand how disaster occurrences change over time.

Power BI was used to create visualizations that illustrate disaster trends across years and months.

Disaster Declarations Over Time

This visualization shows the number of disaster declarations across different years.

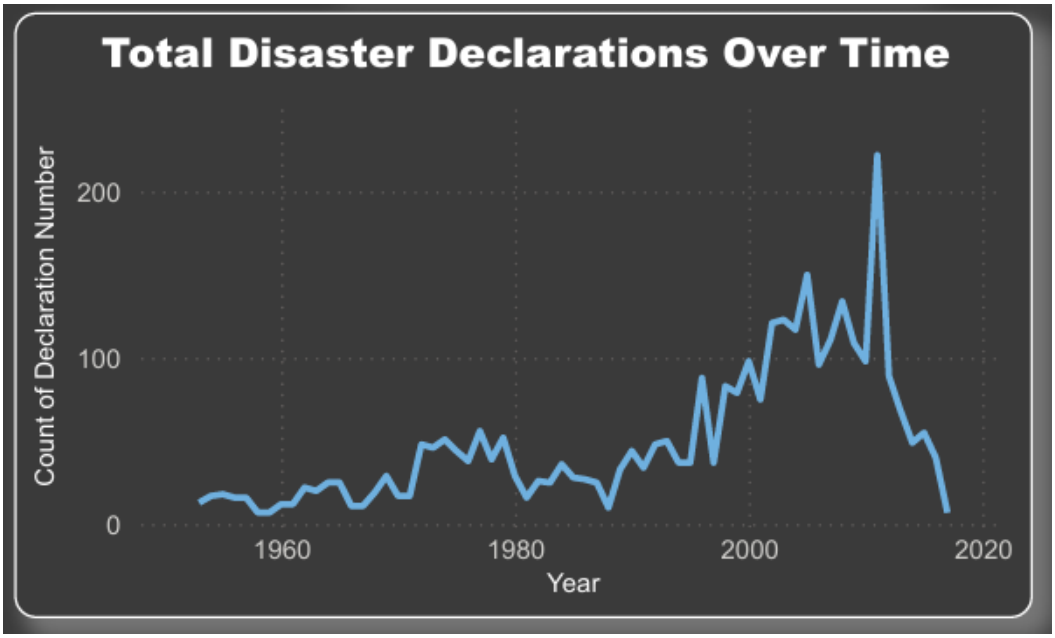


Figure 2: Disaster Declarations Over Time

The chart reveals an increasing trend in disaster declarations over time. This increase may reflect higher disaster frequency, improved reporting systems, or greater awareness of disaster impacts.

Incident Type Trends Over Time

This visualization compares different disaster types across years.

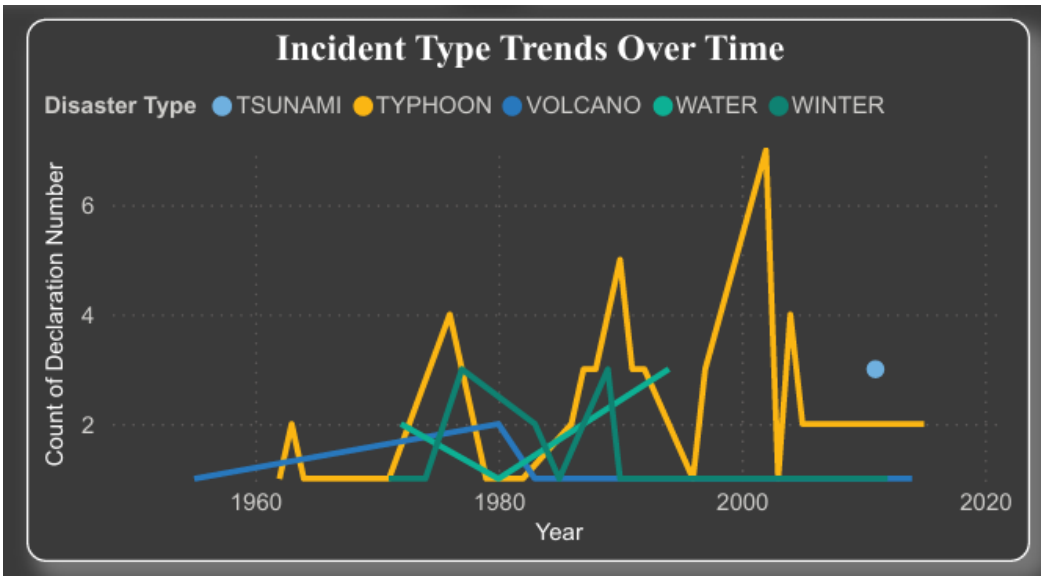


Figure 3: Incident Type Trends Over Time

This chart highlights how certain disaster types have become more frequent over time.

Seasonal Disaster Patterns

Seasonal analysis helps identify months that experience higher disaster activity.

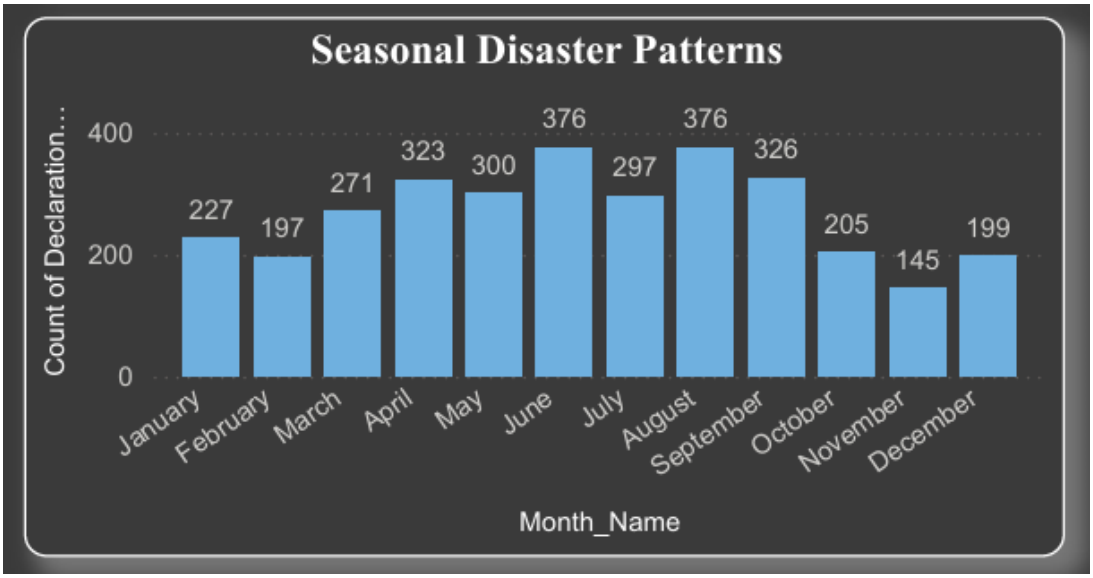


Figure 4: Average Disaster Declarations per Month

The chart indicates that disasters tend to occur more frequently during specific months, which may correspond to seasonal weather patterns.

Disaster Duration Trends

This visualization shows the average duration of disasters across different years.

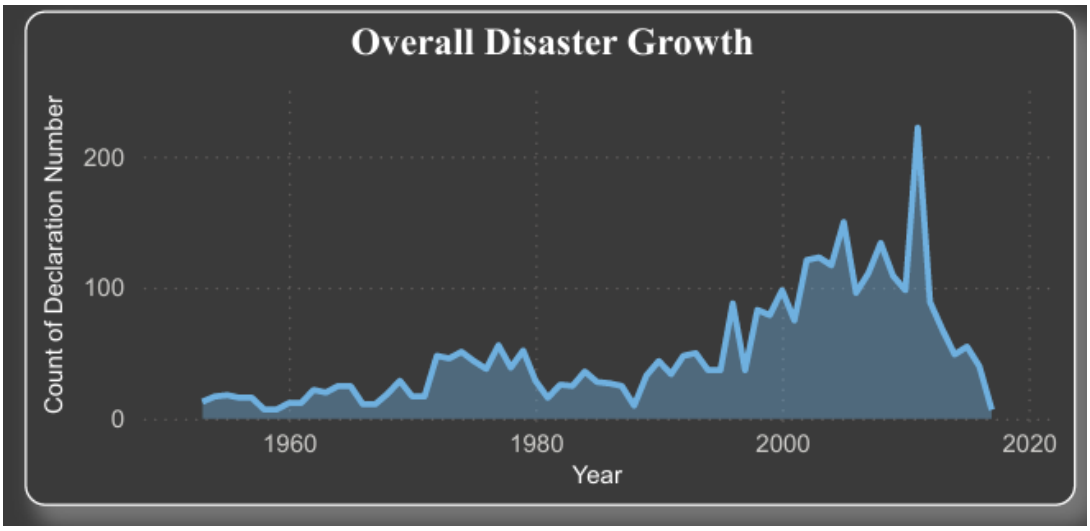
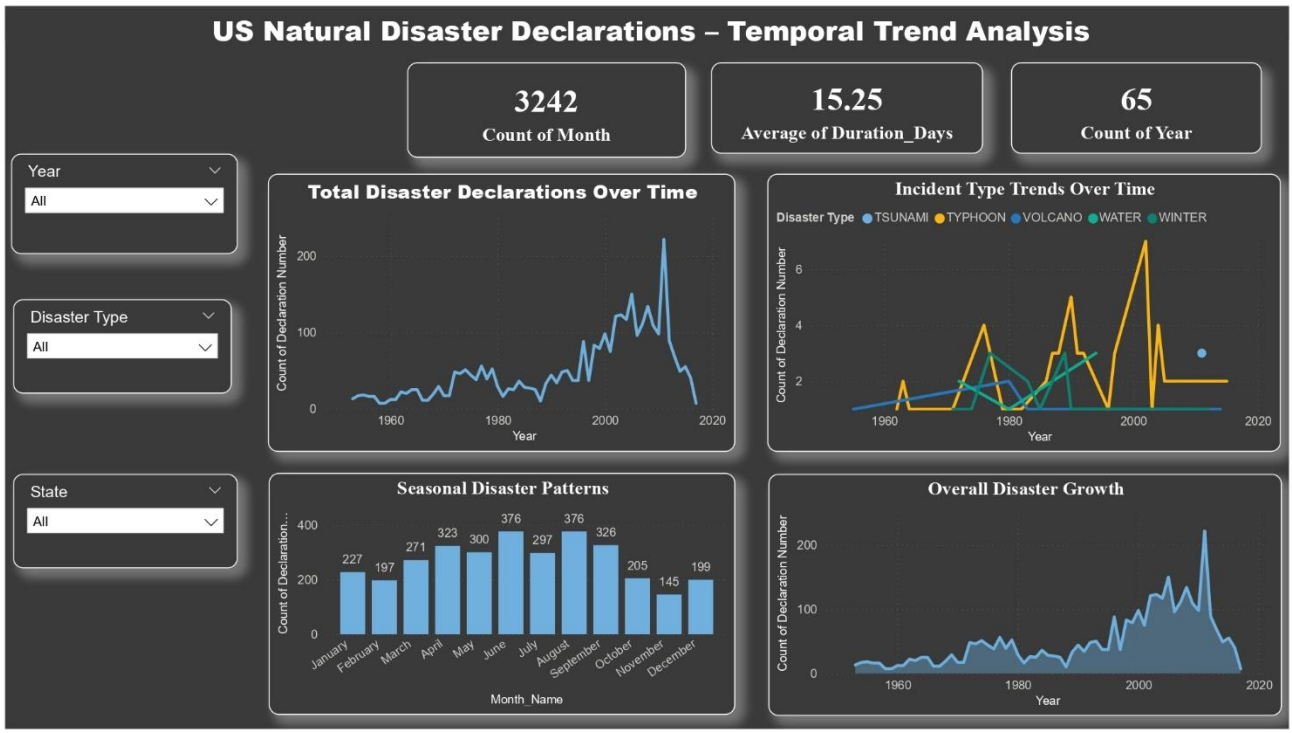


Figure 5: Average Disaster Duration Over Time

This helps understand how long disaster events typically last and whether their duration has changed over time.



6. Geographical Pattern Analysis (Milestone 3)

Geographical analysis was performed to understand how disasters are distributed across different states.

Power BI map visualizations were used to identify regional patterns.

Disaster Declarations by State

This map visualization shows the geographical distribution of disaster declarations across the United States.

Figure 6: Disaster Declarations by State

The map highlights states with higher disaster occurrences, allowing easy identification of disaster-prone regions.

Disaster Intensity Across States

A filled map was used to visualize disaster intensity using color shading.

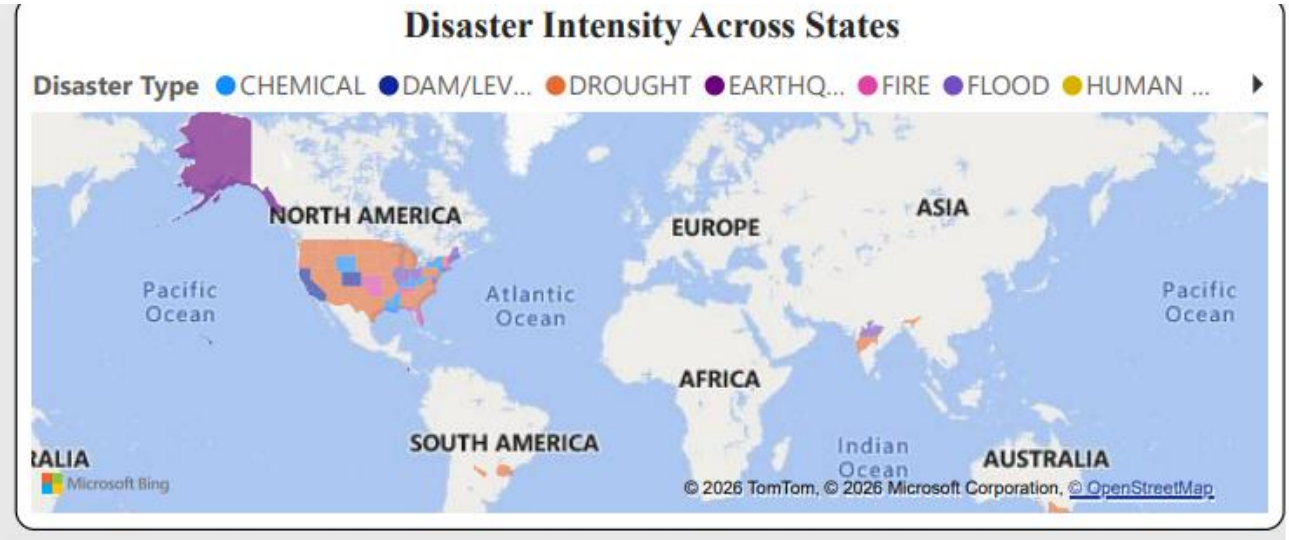


Figure 7: Disaster Intensity Across States

Darker states represent regions with higher disaster declarations.

Top States with Most Disaster Declarations

A bar chart was used to rank the states with the highest number of disaster declarations.

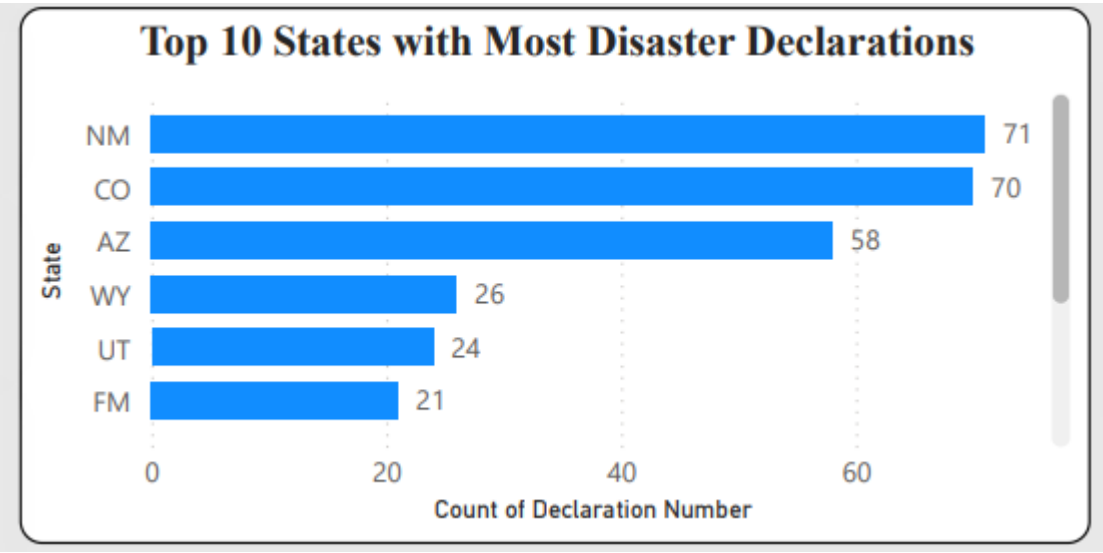


Figure 8: Top States with Most Disaster Declarations

This chart highlights the states most frequently affected by disasters.

Disaster Type Distribution by State

This visualization compares disaster types across different states.

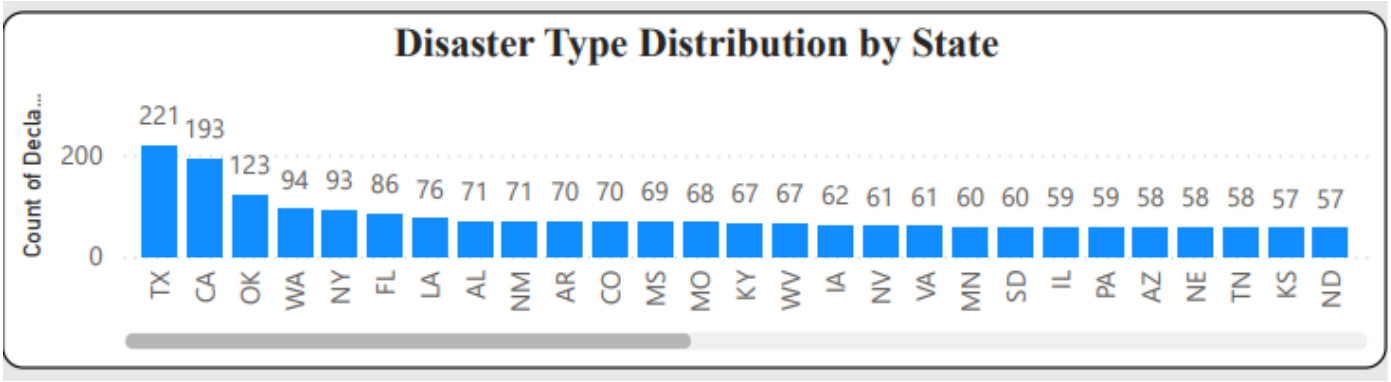


Figure 9: Disaster Type Distribution by State

This helps identify which disasters are more common in certain regions.

Disaster Frequency by State and Type

A matrix heatmap was created to analyze the relationship between states and disaster types.

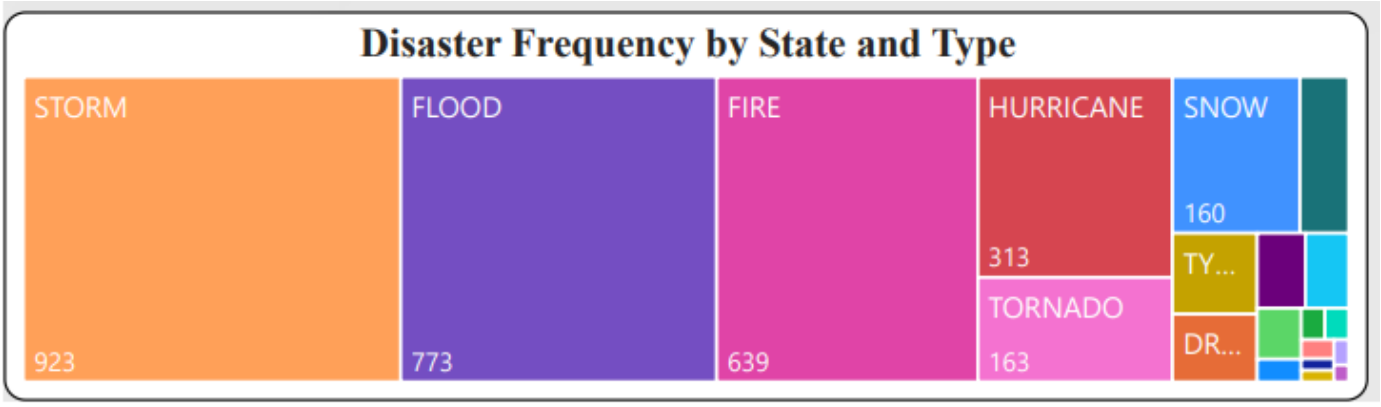


Figure 10: Disaster Frequency by State and Type

This visualization highlights regional patterns in disaster types.

Average Disaster Duration by State

This chart analyzes how disaster duration varies across different states.

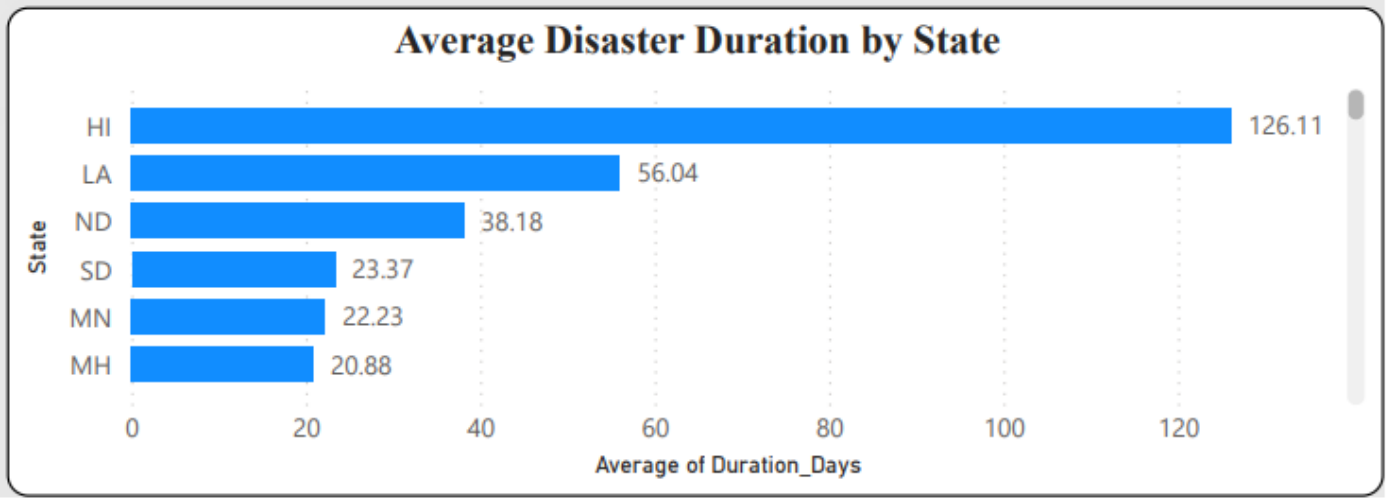


Figure 11: Average Disaster Duration by State

States with longer disaster durations may face greater recovery challenges.

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7. Interactive Dashboard

An interactive Power BI dashboard was created to combine all visualizations into a single analytical interface.

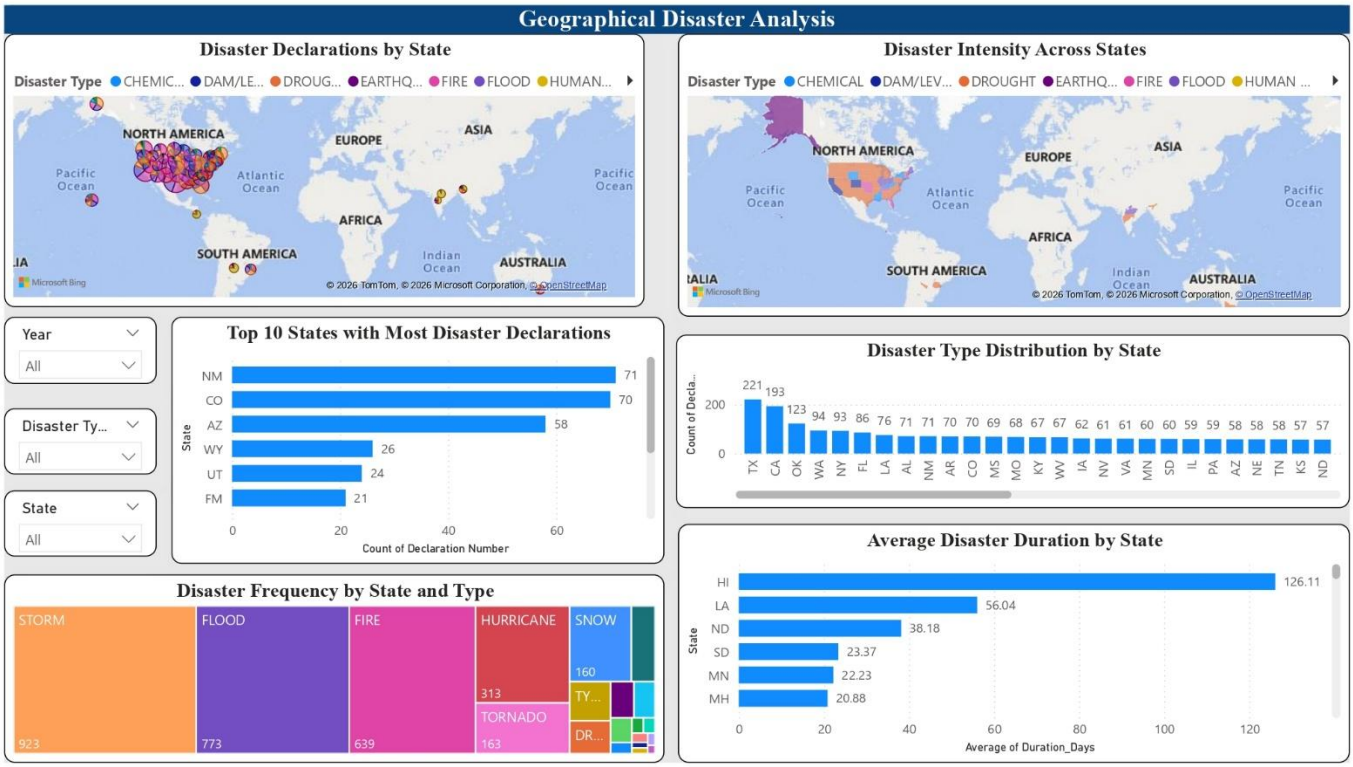
The dashboard includes interactive filters that allow users to dynamically explore the data.

Dashboard Filters

- Year
- Disaster Type
- State

These filters allow users to focus on specific regions, disaster categories, or time periods.

The dashboard provides a user-friendly way to analyze disaster patterns and explore the dataset.



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8. Key Insights

Several important insights were derived from this analysis:

- Disaster declarations have increased significantly over time.
- Certain disaster types have become more frequent in recent decades.
- Seasonal patterns show that disasters tend to occur more often during specific months.
- Some states experience significantly higher disaster frequencies than others.
- Disaster duration varies across states, indicating differences in disaster severity and recovery time.

These insights can help improve disaster preparedness and planning.

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9. Tools and Technologies Used

The following tools were used in this project:

- Python
- Pandas
- Jupyter Notebook
- Power BI

- GitHub

Python was used for data cleaning and preprocessing, while Power BI was used for creating visualizations and dashboards.

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10. Conclusion

This project analyzed disaster declarations in the United States using historical disaster data. Through data cleaning, temporal analysis, and geographical visualization, the project identified important patterns in disaster occurrences.

The analysis shows that disaster occurrences vary across time and geographic regions. Understanding these patterns is important for improving disaster preparedness and response strategies.

Data visualization tools such as Power BI make it possible to transform complex datasets into meaningful insights that can support decision-making.

Future improvements could include predictive modeling to forecast disaster occurrences or integrating additional datasets related to climate conditions and environmental factors.